# The Human Eye and the Colourful World

THE HUMAN EYE AND THE COLOURFUL WORLD

NCERT Class 10 Science - Ch 10 | Hinglish Detailed Notes

NCERT Class 10 Science - Chapter 10 (Hinglish Detailed Notes)

---

COLOUR CODE: RED = Core Basics / extra help (Class 7-8-9 + Ch 9 wali cheezein jo yaad honi chahiye).

COLOUR CODE: GREEN = Board exam me ye topics BAAR-BAAR aate hain - pakka yaad karo.

Baaki normal black text = main chapter content (detail me, simple bhasha me).

## CORE BASICS - PEHLE YE PADHO (CH 9 + FOUNDATION)

---

Ye chapter "Light" (Ch 9) par tika hai. Niche wali base ek baar padh le,
fir Human Eye + dispersion + scattering sab makkhan ho jayega.

1) LENS DO TARAH KE (Ch 9 recap):
- CONVEX (converging) lens = beech me MOTA, kinare patle. Light ko ek point par jodta (converge). Magnifying glass yahi hota.
- CONCAVE (diverging) lens = beech me PATLA, kinare mote. Light ko faila deta (diverge).

2) REFRACTION (apvartan) kya hai:
- Light jab ek medium se dusre (jaise hawa -> kaanch) jaati to apni speed badalti aur MUD (bend) jaati. Isi mudne ko refraction kehte.
- Eye lens aur prism dono refraction se hi kaam karte.

3) FOCUS aur FOCAL LENGTH:
- FOCUS (F) = woh point jahan parallel light rays lens se nikal ke milti hain.
- FOCAL LENGTH (f) = lens ke optical centre se focus tak ki doori.
- f chhoti = lens zyada powerful (zyada mod-ta).

4) REAL vs VIRTUAL IMAGE:
- REAL image = screen par pakdi ja sakti, ULTI (inverted) hoti. (Eye ke retina par bani image REAL + inverted hoti.)
- VIRTUAL image = screen par nahi aati, SEEDHI (erect) hoti (jaise sheeshe me).

5) WHITE LIGHT kya hai:
- Suraj/bulb ki safed roshni asal me 7 colours ka MIXTURE hai. Alag dikhti nahi par prism se alag ho jaati (yahi aage dispersion hai).

6) PRISM kya hai:
- Triangle shape ka transparent kaanch ka tukda (3 rectangular side + 2 triangular side). Isme se light 2 baar refract ho ke mud-ti hai.

7) POWER OF A LENS (dioptre):
- Lens kitna strong hai = uski POWER (P). $P = 1 / f$ (f metre me).
- Unit = DIOPTRE (D). Convex lens ki power +ve, concave ki -ve.
- e.g. $f = 0.5$ m wale convex lens ki $P = +2$ D.

8) NEAR POINT / FAR POINT (idea):
- NEAR POINT = sabse paas ka point jise aankh saaf dekh sakti (normal ~25 cm).
- FAR POINT = sabse door ka point jise saaf dekh sakti (normal = infinity).

# 1. HUMAN EYE - STRUCTURE AUR WORKING

Human eye ek natural camera jaisa hai - light ko refract karke retina par ek REAL aur INVERTED image banata hai. Aankh ke parts + unka kaam exam me bahut puchte hain (diagram + table).

## 1.1 AANKH KE MAIN PARTS AUR UNKA KAAM

CORNEA: aankh ke aage ki transparent (paardarshi) jhilli. Light yahin se andar aati aur SABSE ZYADA refraction (bending) yahi karti hai.
IRIS: cornea ke peeche rangeen (brown/black) part. Pupil ka size control karta aur isi se aankh ka rang dikhta.
PUPIL: iris ke beech ka kaala chhed (hole). Aankh me kitni light jaaye, ye control karta - tej roshni me chhota, andhere me bada ho jaata.
EYE LENS (crystalline lens): jelly jaisa CONVEX lens. Fine focusing karta aur retina par image banata. Iski focal length badal sakti.
CILIARY MUSCLES: eye lens ko pakde rakhte aur uski shape (mota/patla) badal ke focal length change karte (accommodation).
RETINA: aankh ke peeche ki light-sensitive screen (parde jaisi). Yahin REAL + INVERTED image banti. Isme rod aur cone cells hote.
OPTIC NERVE: retina ki image ko electrical signal bana ke DIMAAG (brain) tak pohchata. Dimaag image ko seedha (erect) feel karwata.
AQUEOUS HUMOUR: cornea aur lens ke beech ka patla paani jaisa fluid (shape + nutrition deta).
VITREOUS HUMOUR: lens aur retina ke beech ka gaadha jelly jaisa fluid (aankh ko shape me rakhta).

YAAD RAKHO: image retina par banti hai - REAL aur ULTI (inverted), par brain usse seedha samajhta hai.

## 2. POWER OF ACCOMMODATION

---

DEFINITION: Eye lens ka apni FOCAL LENGTH adjust karne ka power, taaki paas aur door dono cheezein saaf dikhe - isse POWER OF ACCOMMODATION kehte hain.

Kaise hota hai:

- DOOR ki cheez dekhte waqt: ciliary muscles RELAX -> lens PATLA -> focal length BADI -> door ki cheez focus.
- PAAS ki cheez dekhte waqt: ciliary muscles SIKUDTE (contract) -> lens MOTA -> focal length CHHOTI -> paas ki cheez focus.

NEAR POINT (least distance of distinct vision):

- Sabse paas ka point jise aankh bina zor (strain) ke saaf dekh sakti.
- Normal healthy aankh ke liye = 25 cm (isse paas rakho to dhundla dikhega).

### FAR POINT:

- Sabse door ka point jise saaf dekh sakti = INFINITY (normal aankh ke liye).

Lens itna patla/mota ho sakta par ek LIMIT hai - isiliye 25 cm se paas rakhi cheez clearly nahi dikhti (ciliary muscles aur sikud nahi paate).

## 3. DEFECTS OF VISION AUR UNKA CORRECTION

3 main defects: MYOPIA, HYPERMETROPIA, PRESBYOPIA. Har ek ke liye - kya nahi dikhta, image kahan banti, aur kaun sa lens theek karta - ye pakka yaad rakho.

### 3.1 MYOPIA (NEAR-SIGHTEDNESS / nikat-drishti dosh)

- PAAS ki cheez saaf dikhti, par DOOR ki nahi dikhti.
- Image retina ke AAGE (samne, before retina) ban jaati hai.
- FAR POINT infinity se kam ho jaata (door point paas aa jaata).

CAUSE: (i) eye lens ka curvature zyada (focal length bahut chhoti), ya
(ii) eyeball lamba (elongated) ho jana.

CORRECTION: suitable power ka CONCAVE (diverging) lens lagao - ye light ko thoda faila deta taaki image peeche khisak ke retina par bane.

### 3.2 HYPERMETROPIA (FAR-SIGHTEDNESS / door-drishti dosh)

- DOOR ki cheez saaf dikhti, par PAAS ki nahi dikhti.
- Image retina ke PEECHE (beyond retina) banti hai.
- Near point 25 cm se DOOR khisak jaata.

CAUSE: (i) eye lens ka focal length zyada bada, ya
(ii) eyeball chhota (chhota ho jana).

CORRECTION: suitable power ka CONVEX (converging) lens lagao - ye light ko pehle hi thoda jod deta taaki image aage khisak ke retina par bane.

### 3.3 PRESBYOPIA (umar wala dosh)

- Budhape (old age) me aata. Ciliary muscles kamzor + lens ki flexibility kam.
- PAAS aur DOOR dono theek se nahi dikhte (accommodation power ghat jaata).
- Near point door khisak jaata.

CORRECTION: BIFOCAL lens - upar ka hissa CONCAVE (door dekhne ke liye), niche ka hissa CONVEX (paas/padhne ke liye).

### 3.4 CATARACT (motiyabind) - chhota note

- Budhape me kabhi eye lens DHUNDLA/milky (cloudy) ho jaata -> vision kam ya khatam. Ye lens ka defect hai, simple lens se theek nahi hota.
- Ilaaj: CATARACT SURGERY (kharaab lens hata ke artificial lens dalte).

## 4. REFRACTION THROUGH A PRISM

---

Prism ke do refracting surface ek doosre se ek ANGLE par hote - is angle ko PRISM ka ANGLE (A) kehte (aam taur par 60 degree).

- Light jab prism me ghusti to NORMAL ki taraf mud-ti (denser medium), aur jab nikalti to normal se DOOR mud-ti. Do baar refraction hota.

ANGLE OF DEVIATION (D): aane wali ray aur nikalne wali ray ke beech ka angle.
Yani light apne original raaste se kitna mud (deviate) gayi.

Prism me light hamesha uske MOTE hisse (BASE) ki taraf mud-ti hai.
(Kyun? Kyunki base ke paas zyada kaanch hota, light usi taraf jhuk-ti.)

## 5. DISPERSION OF WHITE LIGHT (PRISM SE)

DISPERSION = safed (white) light ka prism se guzar ke apne 7 rangon me TUT jaana. Is rangon ki patti ko SPECTRUM kehte.
7 colours ka order (VIBGYOR): Violet, Indigo, Blue, Green, Yellow, Orange, Red.

KYUN hoti hai dispersion:

- Har colour ki light prism me alag-alag angle se mud-ti hai.
- VIOLET sabse ZYADA mud-ti (bend) hai (uski wavelength sabse chhoti).
- RED sabse KAM mud-ti hai (uski wavelength sabse badi).
- Isiliye colours alag ho ke spectrum ban jaata.

RECOMBINATION (Newton ka 2-prism experiment):

- Newton ne ek prism se white light tod ke spectrum banaya, fir usi spectrum ko ek ULTA (inverted) doosra prism rakh ke wapas jod diya -> dobara WHITE light mil gayi. Isse saabit hua white light = 7 colours ka mixture.

RAINBOW (indradhanush) - natural dispersion:

- Baarish ke baad hawa me latki PAANI ki choti boondein chhote prism jaisa kaam karti.
- Boond me light pehle REFRACT hoti, andar TOTAL INTERNAL REFLECTION se wapas mudti, fir nikalte waqt phir REFRACT + disperse ho jaati.
- Rainbow hamesha suraj ke ULTI taraf (back to the sun) dikhta hai.

## 6. ATMOSPHERIC REFRACTION

Atmosphere (hawa) ki alag-alag parton ka temperature/density alag hota, isliye light bend hoti rehti - isse ATMOSPHERIC REFRACTION kehte. Iske effects:

### 6.1 TWINKLING (TIMTIMANA) OF STARS

- Stars ki light atmosphere ki badalti parton se guzar ke baar-baar thoda-thoda mud-ti (refract) hai, isliye unki position aur brightness halki-halki badalti rehti -> star TIMTIMATA (twinkle) dikhta.
- PLANETS twinkle NAHI karte: woh paas hain to "point" nahi balki choti DISC (bahut saare points) jaise dikhte; sab points ka average effect cancel ho jaata, isliye light steady rehti.

### 6.2 ADVANCE SUNRISE & DELAYED SUNSET

- Atmospheric refraction ki wajah se suraj asli horizon ke niche hote hue bhi dikh jaata.
- Isliye sunrise ~2 minute PEHLE dikh jaata aur sunset ~2 minute BAAD tak dikhta (din thoda lamba feel hota).

### 6.3 SUN NEAR HORIZON - OVAL & REDDISH

- Sunrise/sunset ke waqt suraj horizon ke paas thoda CHAPTA/OVAL aur LAAL-sa dikhta (light ka lamba raasta + refraction). Reddish ka asli reason scattering hai (agla section).

## 7. SCATTERING OF LIGHT

SCATTERING = light ka chhote kann (particles) se takra ke alag-alag dishaon me bikhar jaana. Kitna scatter hoga ye light ke COLOUR (wavelength) par depend karta - chhoti wavelength (BLUE) zyada scatter, badi wavelength (RED) kam.

### 7.1 TYNDALL EFFECT

- Jab light kisi colloid/smoke/dhund (fine particles) me se guzarti to bikhar ke uska RAASTA dikhne lagta - isse TYNDALL EFFECT kehte.
  - Example: ghane jungle me ped ke beech se aati sunlight ki kiran, ya kamre ke chhote chhed se aati roshni jisme dhool ke kann chamakte.

### 7.2 SKY BLUE KYUN DIKHTA HAI

- Hawa ke chhote molecules BLUE (chhoti wavelength) light ko RED se KAHIN ZYADA scatter karte hain.
- Ye bikhri hui neeli light har taraf se aankh me aati -> aasman NEELA dikhta.
  Space me hawa nahi hoti -> scattering nahi -> astronaut ko aasman KAALA dikhta.

### 7.3 SUN SUNRISE/SUNSET PAR LAAL KYUN

- Sunrise/sunset par suraj ki light ko hawa ka BAHUT LAMBA raasta tay karna padta (suraj horizon par hota).
- Lambe raaste me zyadatar BLUE light bikhar (scatter) ke nikal jaati, sirf RED (kam-scatter wali) hum tak pohchti -> suraj LAAL/orange dikhta.
- DOPHAR me suraj sar ke upar, raasta chhota, kam scattering -> safed-sa dikhta.

### 7.4 DANGER / SIGNAL LIGHTS LAAL KYUN

- RED light ki wavelength sabse BADI hoti, isliye woh sabse KAM scatter hoti aur dhund/baarish me bhi bahut DOOR tak saaf dikhti.
- Isiliye STOP signal, brake light, danger sign sab RED rakhe jaate.

## 8. SOLVED EXAMPLES (HARDEST -> EASIEST)

Ye type ke questions exam me pakka aate - reasoning ke saath likhna seekho.

EXAMPLE 1 (Hardest): Myopia aur Hypermetropia me farak - cause, image kahan banti, aur correcting lens, reason ke saath.

MYOPIA: door ki cheez nahi dikhti. Image retina ke AAGE banti. Cause - lens ka curvature zyada YA eyeball lamba. Correction - CONCAVE lens (light faila ke image ko peeche retina par laata).

HYPERMETROPIA: paas ki cheez nahi dikhti. Image retina ke PEECHE banti. Cause - lens ka focal length zyada YA eyeball chhota. Correction - CONVEX lens (light pehle jod ke image ko aage retina par laata).

EXAMPLE 2: Aasman neela aur doobta suraj laal kyun dikhta hai? Poora reason.

Aasman NEELA: hawa ke molecules blue light ko red se zyada scatter karte; ye bikhri neeli light chaaro taraf se aankh me aati -> sky blue.

Suraj LAAL (sunset): light ka lamba raasta, blue scatter ho ke nikal jaati, sirf kam-scatter wali RED light pohchti -> suraj laal/orange.

EXAMPLE 3: Stars timtimate (twinkle) hain par planets kyun nahi?

Stars bahut door, "point source" jaise. Unki light atmosphere ki badalti parton se baar-baar refract hoti -> brightness/position halki badalti -> twinkle.

Planets paas hain, chhoti DISC (bahut points) jaise; sab points ka effect average ho ke cancel ho jaata -> steady light, no twinkle.

EXAMPLE 4: Prism se white light ka dispersion samjhao - violet sabse zyada kyun mud-ta?

White light = 7 colours. Prism me har colour alag angle se mud-ta. VIOLET ki wavelength sabse chhoti to woh sabse ZYADA bend hoti; RED ki wavelength badi to sabse KAM bend. Isliye VIBGYOR spectrum ban jaata.

EXAMPLE 5: Aankh ke parts aur unka kaam - table banao.

CORNEA - light andar laati, sabse zyada refraction.
IRIS - pupil ka size control, aankh ka rang.
PUPIL - andar jaane wali light ki maatra control.
LENS - focal length badal ke retina par image banata.
RETINA - light-sensitive screen, real+inverted image yahin banti.
OPTIC NERVE - signal brain tak le jaata.

EXAMPLE 6: Power of accommodation kya hai aur normal aankh ka near point kitna?

Eye lens ka focal length badal ke paas-door dono focus karne ka power = power of accommodation. Ciliary muscles lens ko mota/patla karte. Normal near point = 25 cm, far point = infinity.

EXAMPLE 7 (Easiest): Danger signal aur traffic stop light laal kyun hoti?

RED ki wavelength sabse badi -> sabse kam scatter -> dhund me bhi door tak saaf dikhti, isliye danger/stop signal red rakhte.

## 9. QUICK Q&A - PADHNE KE BAAD KHUD CHECK KARO

Pehle khud answer socho, fir niche milao. Ye board-style questions hain.

Q1. Retina par bani image kaisi hoti hai?
A1. REAL aur INVERTED (ulti); brain usse seedha samjha deta.

Q2. Pupil ka kya kaam hai aur use control kaun karta?
A2. Aankh me jaane wali light ki maatra control karta; IRIS use chhota/bada karta.

Q3. Least distance of distinct vision (near point) normal aankh ke liye kitna?
A3. 25 cm. Far point = infinity.

Q4. Myopia me image kahan banti aur kaun sa lens lagta?
A4. Retina ke AAGE; CONCAVE (diverging) lens se theek hota.

Q5. Hypermetropia me image kahan banti aur kaun sa lens lagta?
A5. Retina ke PEECHE; CONVEX (converging) lens se theek hota.

Q6. Presbyopia kaun se lens se theek karte?
A6. BIFOCAL lens se (upar concave - door, niche convex - paas).

Q7. Dispersion me sabse zyada aur sabse kam koun sa colour mud-ta?
A7. Sabse zyada VIOLET, sabse kam RED.

Q8. Stars twinkle karte par planets nahi - ek line ka reason?
A8. Star point source hai (light fluctuate hoti); planet disc jaisa hai (effect average ho ke cancel).

Q9. Tyndall effect kya hai?
A9. Fine particles (colloid/dhund/dhool) se light ka scatter ho ke uska raasta dikhna.

Q10. Sunset par suraj laal kyun? Ek line.
A10. Lambe raaste me blue scatter ho jaati, sirf kam-scatter wali RED light pohchti, isliye suraj laal dikhta.

---

EXAM STRATEGY (1 minute revision):

- Aankh ke parts + kaam (cornea, iris, pupil, lens, ciliary, retina) ka diagram + table - har saal aata.
- Power of accommodation + near point (25 cm) / far point (infinity).
- 3 defects: MYOPIA (concave), HYPERMETROPIA (convex), PRESBYOPIA (bifocal) - image kahan banti + lens, ratlo.
- Dispersion + VIBGYOR (violet most, red least) + recombination + rainbow.
- Atmospheric refraction: twinkling, planets no-twinkle, advance sunrise.
- Scattering: sky BLUE, sun RED at sunset, danger signal RED, Tyndall effect.
- CORE BASICS page (lens, refraction, focus, dioptre, real/virtual image) bhool jaao to wapas pehle padho - base strong rakho.

---

Generated by Kiro for Krishna1k - All the best!